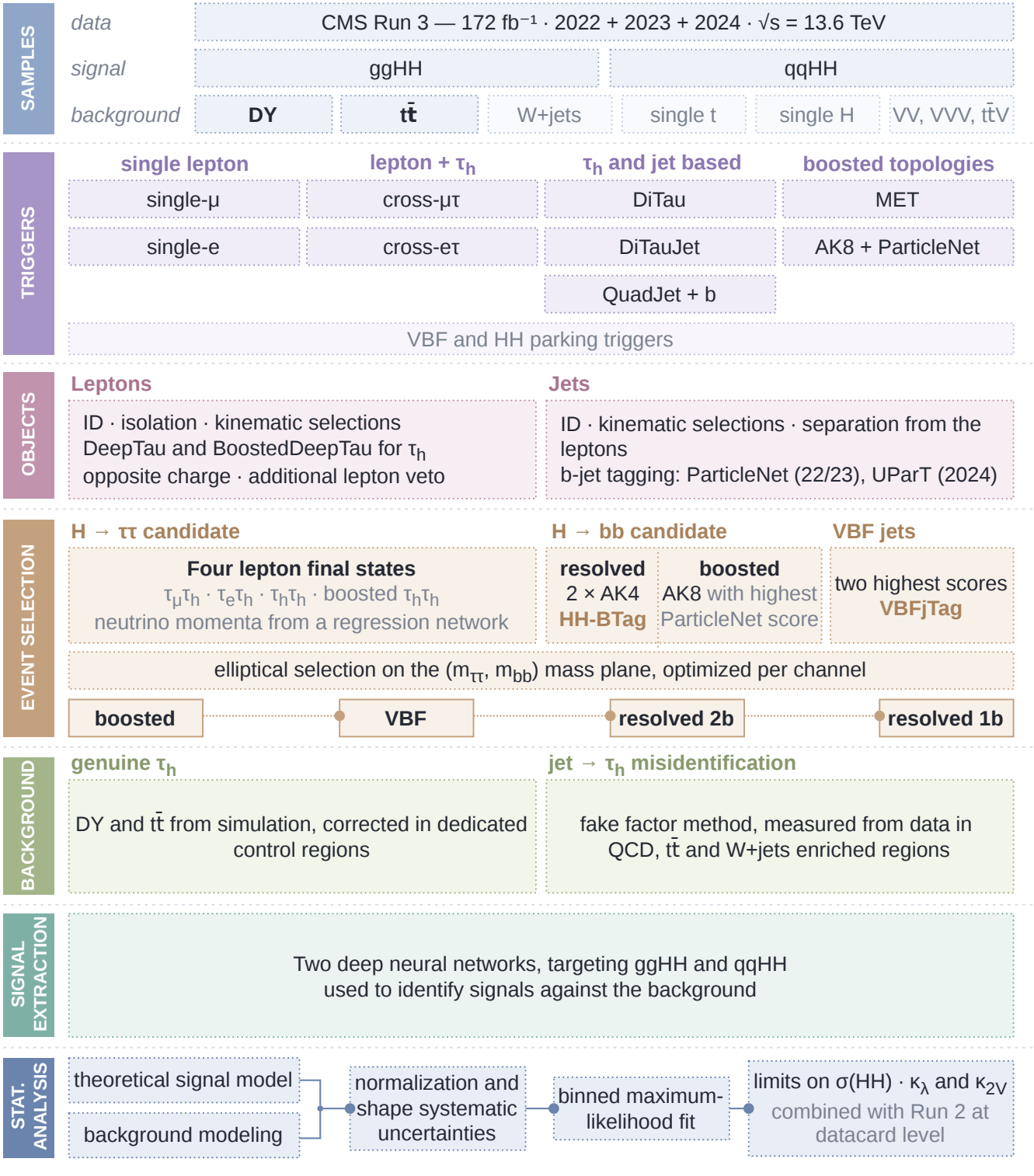

SAMPLES
data
CMS Run 3 — 172 fb⁻¹ · 2022 + 2023 + 2024 · √s = 13.6 TeV
signal
ggHH
qqHH
background
DY
tt̄
W+jets
single t
single H
VV, VVV, tt̄V
TRIGGERS
single lepton
lepton + τh
τh and jet based
boosted topologies
single-μ
cross-μτ
DiTau
MET
single-e
cross-eτ
DiTauJet
AK8 + ParticleNet
QuadJet + b
VBF and HH parking triggers
OBJECTS
Leptons
ID · isolation · kinematic selections
DeepTau and BoostedDeepTau for τh
opposite charge · additional lepton veto
Jets
ID · kinematic selections · separation from the leptons
b-jet tagging: ParticleNet (22/23), UParT (2024)
EVENT SELECTION
H → ττ candidate
Four lepton final states
τμτh · τeτh · τhτh · boosted τhτh
neutrino momenta from a regression network
H → bb candidate
resolved
2 × AK4
HH-BTag
boosted
AK8 with highest ParticleNet score
VBF jets
two highest scores
VBFjTag
elliptical selection on the (mττ, mbb) mass plane, optimized per channel
boosted
VBF
resolved 2b
resolved 1b
BACKGROUND
genuine τh
DY and tt̄ from simulation, corrected in dedicated control regions
jet → τh misidentification
fake factor method, measured from data in QCD, tt̄ and W+jets enriched regions
SIGNAL EXTRACTION
Two deep neural networks, targeting ggHH and qqHH
used to identify signals against the background
STAT. ANALYSIS
theoretical signal model
background modeling
normalization and shape systematic uncertainties
binned maximum-likelihood fit
limits on σ(HH) · κλ and κ2V
combined with Run 2 at datacard level